$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\end{pmatrix} \begin{pmatrix}2\\1\end{pmatrix} \begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}5\\4\end{pmatrix} \begin{pmatrix}6\\5\end{pmatrix} \begin{pmatrix}6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

Initial $\boldsymbol{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ $Step\,\eta = 1$

Check classes of $Z_1 \; Z_2 \; Z_3$ $\begin{pmatrix}0\\0\end{pmatrix} \begin{pmatrix}7\\6\end{pmatrix} \begin{pmatrix}3\\2\end{pmatrix}$

C_1 *positive* C_2 *negative* ***Zero*** is unaccepted

Ugmented Vectors (from Top)

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\\1\end{pmatrix} \begin{pmatrix}1\\2\\1\end{pmatrix} \begin{pmatrix}1\\1\\2\end{pmatrix} \begin{pmatrix}1\\5\\4\end{pmatrix} \begin{pmatrix}1\\6\\5\end{pmatrix} \begin{pmatrix}1\\6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

Check Classes

$d(Z_1) = \begin{pmatrix}1 & 0 & 0\end{pmatrix} \begin{pmatrix}5\\-1\\-1\end{pmatrix} = 5$ ⇒ Positive > 0 **Class I**

$d(Z_2) = \begin{pmatrix}1 & 7 & 6\end{pmatrix} \begin{pmatrix}5\\-1\\-1\end{pmatrix} = -8$ ⇒ Negative < 0 **Class II**

$d(Z_3) = \begin{pmatrix}1 & 3 & 2\end{pmatrix} \begin{pmatrix}5\\-1\\-1\end{pmatrix} = 0$ ⇒ Not > 0 , Not < 0 **Undefined Class**